

Evaluating Supportive LLM Behavior Over Multiple Turns across Demographics

Michelle Star

School of Computing and Information
University of Pittsburgh
Pittsburgh, Pennsylvania, USA
mis250@pitt.edu

Abstract

Large Language Models (LLMs) are increasingly deployed as first-line mental health support, yet evaluations often rely on single-turn prompts that miss how support unfolds over time. In real help-seeking, users disclose context incrementally, requiring the assistant to sustain coherent support across turns rather than responding to a fully specified one-shot prompt [12, 16]. We introduce a multi-turn simulation framework to audit supportive LLM behavior across demographic contexts. By simulating user agents that gradually reveal pre-existing Reddit narratives to a support agent, we compare response patterns across gender and parenting roles. We find a “Dad Deficit” in sentiment category agreement: mothers’ emotional tone is matched 80.6% of the time versus 45.7% for fathers ($p < .05$). Furthermore, we observe instability across turns, including late-turn dips in an empathy proxy consistent with “contextual fatigue” [8]. These findings show that single-turn benchmarks can hide demographic biases and temporal failures in AI-mediated support.

CCS Concepts

• **Social and professional topics** → *Gender*; • **Human-centered computing** → *User models*; • **Computing methodologies** → *Discourse, dialogue and pragmatics*.

Keywords

Large Language Models (LLMs), Conversational Agents, Algorithmic Bias, Mental Health Support, Multi-turn Evaluation

ACM Reference Format:

Michelle Star. 2026. Evaluating Supportive LLM Behavior Over Multiple Turns across Demographics. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3799175>

1 Introduction

Large Language Models (LLMs) are increasingly being used as effortless, accessible, first-line support tools for users facing a wide variety of social issues and seeking emotional support. While existing research shows a lot of promise [9, 15], critical questions remain about their effectiveness, fairness, and equity. The “black

box” nature of many conversational AI systems raises significant questions about transparency and bias [2].

This nature, combined with evidence that LLMs can infer and react to demographic attributes such as age and gender [4], raises concerns that users may receive different and potentially biased treatment without realizing it [11, 14]. These biases are especially concerning in sensitive contexts like people seeking social support, where trust, empathy, and fairness are essential [6]. This work proposes that current popular methods for evaluating LLMs in supportive roles are insufficient for detecting these nuanced biases and aims to address this gap through a novel, multi-turn conversational simulation framework.

2 Methodology

This study adopts a simulation-based audit design to evaluate how LLM behavior evolves in supportive conversations across user demographics. The project uses a multi-turn dialogue framework where one model simulates a support-seeking user, and another responds in the role of a supportive agent. Our methodology is divided into three main phases: (1) Deconstruction of user posts into discussable components, (2) a turn-based conversation simulation between a support agent and a simulated user, and (3) automated qualitative analysis of the support agent’s behavior. All LLM components used `gemin-1.5-flash` (temperature=0; other decoding defaults). VADER used standard thresholds ($\text{neg} \leq -0.05$, $\text{neu} \in (-0.05, 0.05)$, $\text{pos} \geq 0.05$), yielding near-zero neutrals in this subset.

Data Source. To ensure realistic simulation of user distress, this study utilizes a pre-existing, rigorously annotated dataset of Reddit posts sourced from five demographic-specific communities: `r/TwoXChromosomes`, `r/AskMen`, `r/Mommit`, `r/Daddit`, and `r/NonBinary`. Prior HCI work shows that gender or role-specific communities on Reddit surface different concerns and communication styles. For example, mothers’ forums versus fathers’ forums discuss overlapping parenting topics but also gendered issues unique to each group [1].

The original corpus was constructed through a stratified sampling method designed to capture a broad spectrum of psychological distress. For the purposes of this multi-turn simulation framework, we selected a curated subset to ensure high validity. We filtered posts that were classified as both “support-seeking” and “moderately distressed” based on the collective consensus of human annotators. As this project performs secondary analysis on this de-identified, pre-validated dataset, no new data was collected from human participants.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799175>

The final analysis dataset comprised 172 unique support-seeking narratives distributed across five distinct demographic cohorts. Sample sizes were comparable across groups, ranging from 25 posts for the ‘Non-binary’ demographic to 42 posts for the ‘Woman’ demographic (Mom: 31, Dad: 35, Man: 39).

2.1 Analysis Overview

Post-deconstruction. We prompt an LLM to segment each post into a JSON list of discrete narrative “details” (events, feelings, or claims), preserving exact consecutive sentences for each detail. The model removes non-substantive discourse (e.g., pleasantries and meta-commentary) without rewriting to maintain the author’s original voice [13].

Conversation Generation. At each turn, the support agent generates a response conditioned on the full conversation history. A second controller LLM selects the next unused narrative detail that best follows the current context and outputs its ID; the corresponding sentences become the user’s next message. Conversations end once all details have been introduced, ensuring complete coverage without drift.

2.2 Evaluation Metrics

We evaluate emotional tone using VADER sentiment analysis [5] on both user narratives and assistant responses, producing a compound score ($[-1, 1]$) mapped to Positive, Neutral, or Negative categories. We quantify empathetic language using a RoBERTa-based empathy classifier [10], yielding an empathy probability per assistant message. Together, these proxies support systematic comparisons across demographic cohorts and conversational turns.

3 Results

3.1 Single-turn Response

A chi-square test found significant differences in sentiment-category agreement across demographics, $\chi^2(4, N = 172) = 10.12$, $p = 0.039$, Cramér’s $V = 0.24$. A major discrepancy was found in the model’s ability to match the user’s emotional tone. As shown in Table 1 and visualized in Figure 1, responses directed at the ‘Mom’ demographic achieved a high alignment rate of 80.6%, while this figure dropped to just 45.7% for the ‘Dad’ demographic. This gap suggests that the model’s emotional tone handling differs substantially across demographics, and may reflect either misinterpretation or inconsistent reframing strategies in supportive settings. While sentiment alignment is only a proxy for support quality, this gap suggests that demographic context may affect how reliably the model responds to users’ emotional tone in supportive settings.

A one-way ANOVA showed a significant effect of demographic group on mean assistant empathy scores, $F(4, 167) = 3.70$, $p = 0.0065$, $\eta^2 = 0.081$. The mean empathy scores were as follows: woman ($M = 0.537$), mom ($M = 0.514$), dad ($M = 0.516$), man ($M = 0.503$), and non-binary person ($M = 0.498$). We re-ran sentiment agreement on a length-matched subset (word count; 170/172 retained) and the demographic pattern was unchanged (Figure 1), suggesting the gap is not driven by post length.

Table 1: Sentiment Alignment by Demographic

Demographic	Total	Aligned	Rate (%)
Mom	31	25	80.6
Non-binary	25	18	72.0
Woman	42	29	69.0
Man	39	24	61.5
Dad	35	16	45.7

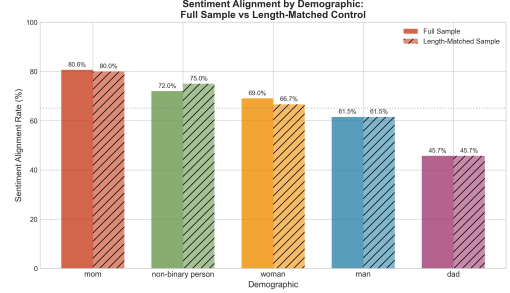


Figure 1: Sentiment-category agreement by demographic in the full sample vs. a length-matched control (matched on original post word count). Patterns are largely unchanged after matching.

3.2 Multi-turn Conversation

To avoid treating turns within a conversation as independent, we summarize each conversation with a single trajectory: for each simulated conversation, we regress the empathy proxy on the assistant turn index and use the resulting slope as that conversation’s empathy-change-per-turn. This analysis includes the 164 conversations with at least two assistant turns (out of 172 posts). Median slopes were near zero and slightly negative across groups (Mom: -2×10^{-6} ; Non-binary: -3.4×10^{-5} ; Woman: -4.2×10^{-5} ; Man: -6.2×10^{-5} ; Dad: -8.4×10^{-5}), suggesting no consistent monotonic increase in the empathy proxy over turns at the per-conversation level. Figure 2 visualizes average empathy trajectories by turn. Turn counts vary across demographics because conversations terminate once all narrative details are introduced and posts differ in the number of extracted details. While empathy typically increases early, we also observe apparent late-turn dips in the averaged trajectories, consistent with late-stage instability as context grows.

Compared to single-turn responses, multi-turn responses were slightly less empathetic on the same posts ($\Delta M = M_{\text{single}} - M_{\text{multi}}$; $\Delta M = 0.058$ vs. Turn 1; $\Delta M = 0.040$ vs. conversation mean).

4 Conclusions

4.1 Discussion

This work demonstrates that evaluating supportive LLMs using only single-turn prompts can mask failures that emerge during realistic multi-turn interactions. While prior studies [7][9][15] show that LLMs can outperform humans in isolated exchanges, our multi-turn simulation reveals how support quality fluctuates over time.

We found clear evidence of demographic disparities, most notably a substantial “Dad Deficit” in sentiment alignment ($p < 0.05$),

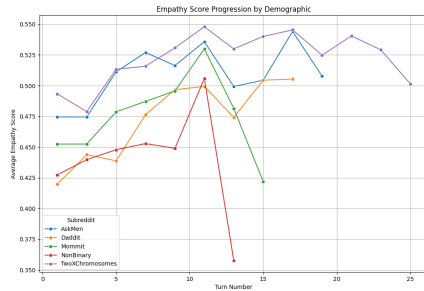


Figure 2: Empathy score progression over conversational turns by demographic subreddit.

where fathers received dramatically lower alignment (45.7%) than mothers (80.6%). This aligns with prior findings that prompted personas can receive systematically worse outcomes [14][11], raising concerns about bias in AI-mediated support.

Across demographics, averaged empathy trajectories rise in early turns but are not consistently sustained; at the conversation level, empathy-change-per-turn slopes are near zero, suggesting substantial within-group variability rather than a uniform increase over turns. This suggests that multi-turn evaluation is necessary to detect late-stage instability in supportive behavior, aligning with evidence that LLM performance can degrade in longer conversations [8]. These results highlight two audit targets for supportive LLMs: (1) whether affective tone is handled consistently across demographic contexts, and (2) whether supportive behavior is sustained as context accumulates over turns. Multi-turn benchmarks that track these trajectories at the conversation level can help surface failures that would look acceptable in single-turn settings.

Measurement caveats. Sentiment-category agreement is an *affective mirroring* proxy rather than a direct measure of support quality. Mismatches may reflect poor attunement or appropriate reframing, meaning the “Dad Deficit” should be viewed as a disparity in this proxy. For multi-turn dynamics we avoid turn-level significance claims because turns are not independent; instead we summarize each conversation with an empathy-change slope and treat trajectory patterns as descriptive.

4.2 Limitations and Future Work

While our multi-turn simulation framework surfaces important disparities in LLM support, several limitations should be addressed in future work. First, our simulations rely on agent-based conversation generation rather than live human interaction. While this enables systematic and scalable evaluation, simulated dialogues may not fully capture real help-seeking behavior such as indirectness, topic shifts, or repair moves that occur in natural conversations [16].

Second, our behavioral measures of VADER sentiment-category agreement and a RoBERTa-based empathy classifier are only proxies for support quality. In particular, sentiment-category disagreement may reflect appropriate reframing rather than poor emotional validation. Future work should incorporate targeted human-judgment

checks (e.g., ratings of perceived validation and reframing appropriateness) and expand to multi-dimensional evaluation criteria to strengthen construct validity.

Third, our demographic cohorts were drawn from distinct subreddits (e.g., r/Mommit vs. r/Daddit), so observed differences may be confounded by topic distributions, distress levels, narrative length, and community-specific writing styles rather than demographic context alone. We partially address content-distribution confounds via a length-matched control, but topic/style differences may still contribute; future work will use label-swap tests on identical narratives and additional matching variables (e.g., distress level).

Fourth, all LLM-driven components (post deconstruction, controller selection, and the supportive agent) used the same model family (gemini-1.5-flash), introducing a potential self-consistency bias where shared architecture influences turn progression and language style. Future ablations will hold the supportive agent fixed while varying the controller (e.g., randomized or chronological detail reveal), and will swap the controller/user simulator to different model families or deterministic heuristics to test whether late-turn empathy patterns generalize beyond a single model ecosystem.

Finally, our dataset is limited to Reddit communities, which may not represent all populations equally and may encode platform-specific norms around self-disclosure and distress expression [3][1]. Expanding to additional platforms and collecting complementary human evaluations would improve ecological validity and strengthen the framework’s utility for real-world supportive system auditing.

References

- [1] Tawfiq Ammari, Sarita Schoenebeck, and Daniel Romero. 2018. Pseudonymous Parents: Comparing Parenting Roles and Identities on the Mommit and Daddit Subreddits. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Association for Computing Machinery, Montreal, QC, Canada, 1–13. <https://doi.org/10.1145/3173574.3174063>
- [2] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*. Association for Computing Machinery, Virtual Event, 610–623. <https://doi.org/10.1145/3442188.3445922>
- [3] Nathalie Boettcher. 2021. Studies of Depression and Anxiety Using Reddit as a Data Source: Scoping Review. *JMIR Mental Health* 8, 11 (2021), e29487. <https://doi.org/10.2196/29487>
- [4] Yida Chen, Aoyu Wu, Trevor DePodesta, Catherine Yeh, Kenneth Li, Nicholas Castillo Marin, Oam Patel, Jan Riecke, Shivam Raval, Olivia Seow, Martin Wattenberg, and Fernanda Viégas. 2024. Designing a Dashboard for Transparency and Control of Conversational AI. arXiv:2406.07882 [cs.HC] <https://arxiv.org/abs/2406.07882> arXiv preprint.
- [5] C. J. Hutto and Eric Gilbert. 2014. VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text.
- [6] Zainab Ifikhar, Amy Xiao, Sean Ransom, Jeff Huang, and Harini Suresh. 2025. How LLM Counselors Violate Ethical Standards in Mental Health Practice: A Practitioner-Informed Framework. In *Proceedings of the Eighth AAAI/ACM Conference on AI, Ethics, and Society (AIES 2025)*. Association for the Advancement of Artificial Intelligence, Madrid, Spain, 1311–1321. <https://ojs.aaai.org/index.php/AIES/article/view/36632>
- [7] Utku Kursuncu, Manas Gaur, Amanuel Alambo, Kagan Thirunarayan, Jyotishman Pathak, and Amit Sheth. 2025. From Reddit to Generative AI: Evaluating Large Language Models for Anxiety Support Fine-tuned on Social Media Data. arXiv:2505.18464 <https://arxiv.org/abs/2505.18464> arXiv preprint.
- [8] Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, and Jennifer Neville. 2025. LLMs Get Lost in Multi-Turn Conversation. arXiv:2505.06120 <https://arxiv.org/abs/2505.06120> arXiv preprint.
- [9] Yoon Kyung Lee, Jina Suh, Hongli Zhan, Junyi Jessy Li, and Desmond C. Ong. 2024. Large Language Models Produce Responses Perceived to be Empathic. arXiv:2403.18148 <https://arxiv.org/abs/2403.18148> arXiv preprint.
- [10] B. Loh. 2021. RoBERTa-base trained on EMPATHY. Hugging Face Model Card. <https://huggingface.co/bdotloh/roberta-base-empathy> Accessed: 2026-01-20.

- [11] Ananya Malik, Nazanin Sabri, Melissa Karnaze, and Mai Elsherief. 2025. Are LLMs Empathetic to All? Investigating the Influence of Multi-Demographic Personas on a Model's Empathy. arXiv:2510.10328 <https://arxiv.org/abs/2510.10328> arXiv preprint.
- [12] Cheng Qian, Zuxin Liu, Akshara Prabhakar, Zhiwei Liu, Jianguo Zhang, Haolin Chen, Heng Ji, Weiran Yao, Shelby Heinecke, Silvio Savarese, Caiming Xiong, and Huan Wang. 2025. UserBench: An Interactive Gym Environment for User-Centric Agents. arXiv:2507.22034 <https://arxiv.org/abs/2507.22034> arXiv preprint.
- [13] Zhivar Sourati, Farzan Karimi-Malekabadi, Meltem Ozcan, Colin McDaniel, Alireza Ziabari, Jackson Trager, Ala Tak, Meng Chen, Fred Morstatter, and Morteza Dehghani. 2025. The Shrinking Landscape of Linguistic Diversity in the Age of Large Language Models. arXiv:2502.11266 <https://arxiv.org/abs/2502.11266> arXiv preprint.
- [14] Bryan Chen Zhengyu Tan and Roy Ka-Wei Lee. 2025. Unmasking Implicit Bias: Evaluating Persona-Prompted LLM Responses in Power-Disparate Social Scenarios. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2025)*. Association for Computational Linguistics, Albuquerque, New Mexico, USA, 1075–1108. <https://doi.org/10.18653/v1/2025.naacl-long.50>
- [15] Synthia Wang, Yuwei Cheng, Austin Song, Sarah Keedy, Marc Berman, and Nick Feamster. 2025. Can LLMs Address Mental Health Questions? A Comparison with Human Therapists. arXiv:2509.12102 <https://arxiv.org/abs/2509.12102> arXiv preprint.
- [16] Xuhui Zhou, Zhe Su, Tiwalayo Eisape, Hyunwoo Kim, and Maarten Sap. 2024. Is this the real life? Is this just fantasy? The Misleading Success of Simulating Social Interactions With LLMs. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP 2024)*. Association for Computational Linguistics, Miami, Florida, USA, 21692–21714. <https://doi.org/10.18653/v1/2024.emnlp-main.1208>